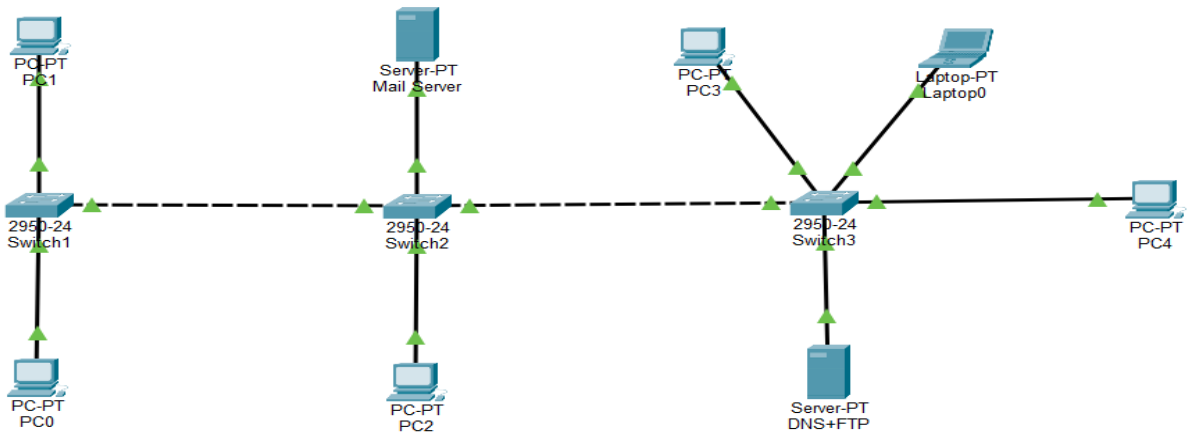
Computer Networks Lab 6

Muhammad Saad Tahir

24k-0851 BCS-5B

TASK1:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(ed...	(delete)
	Successful	PC2	Server0	ICMP		0.000	N	1	(ed...	(delete)
	Successful	PC3	PC4	ICMP		0.000	N	2	(ed...	(delete)
	Successful	Lapto...	Server1	ICMP		0.000	N	3	(ed...	(delete)

S

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 09.56.1.1 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

S

```
Switch(config-if)#exit
Switch(config)#line vty 0 4
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#transport input telnet
Switch(config-line)#exit
Switch(config)#
```

S

```
Switch>en
Switch#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 09.56.2.1 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

```
Switch(config-if)#exit
Switch(config)#line vty 0 4
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#transport input telnet
Switch(config-line)#exit
Switch(config)#
```

```
Switch>en
Switch#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip address 09.56.3.1 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

```
Switch(config-if)#exit
Switch(config)#line vty 0 4
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#transport input telnet
Switch(config-line)#exit
Switch(config)#
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 09.56.1.1
Trying 9.56.1.1 ...Open


User Access Verification

Password:
Password:
Switch>en
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 09.56.2.1
Trying 9.56.2.1 ...Open


User Access Verification

Password:
Switch>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 09.56.3.1
Trying 9.56.3.1 ...Open


User Access Verification

Password:
Switch>
```

0	ftp.saad.com	A Record	9.56.3.20
1	mail.saad.com	A Record	9.56.2.20

a

1	cisco	cisco	RWDNL
2	saad	123	RWDNL

s

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 09.56.3.20
Trying to connect...09.56.3.20
Connected to 09.56.3.20
220- Welcome to PT Ftp server
Username:saad
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>
```

s

```
C:\>ping ftp.fasih.com

Pinging 9.56.3.20 with 32 bytes of data:

Reply from 9.56.3.20: bytes=32 time=1ms TTL=128
Reply from 9.56.3.20: bytes=32 time<1ms TTL=128
Reply from 9.56.3.20: bytes=32 time<1ms TTL=128
Reply from 9.56.3.20: bytes=32 time<1ms TTL=128

Ping statistics for 9.56.3.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PC2

Physical Config Desktop Programming Attributes

Configure Mail X

User Information

Your Name: saad

Email Address: saad@saad.com

Server Information

Incoming Mail Server: 09.56.2.20

Outgoing Mail Server: 09.56.2.20

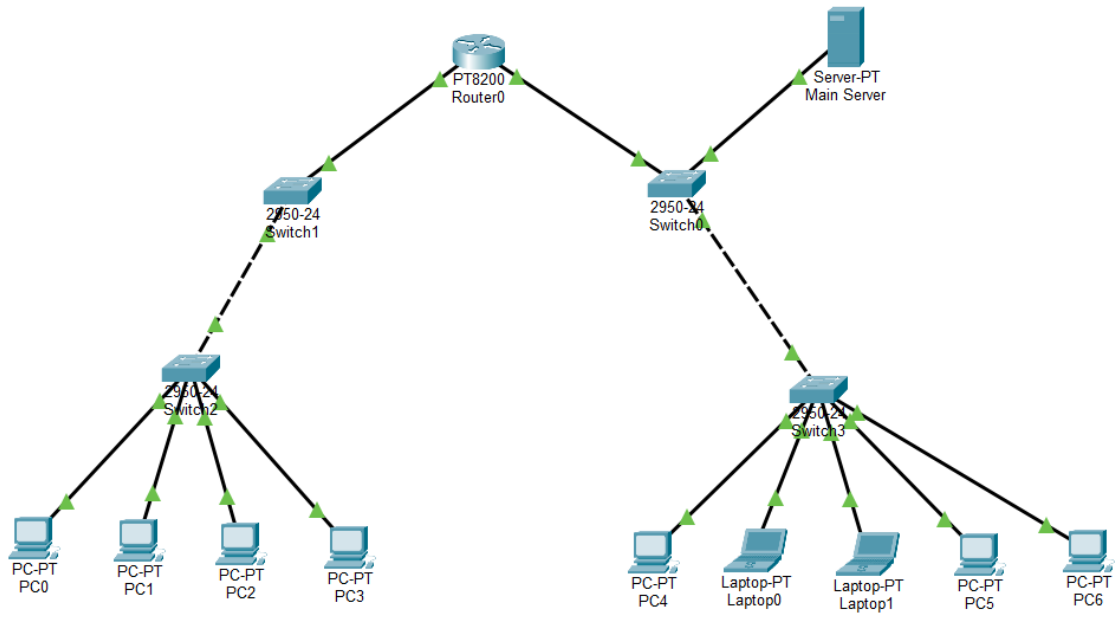
Logon Information

User Name: saad

Password: ••••

Save Remove Clear Raset

TASK2:



S

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC0	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC4	ICMP		0.000	N	2	(edit)	(delete)
	Failed	PC0	Main Server	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC0	Main Server	ICMP		0.000	N	4	(edit)	(delete)

c

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
%Invalid interface type and number
Router(config)#interface Fa0/0
%Invalid interface type and number
Router(config)#interface Fa 0/0
%Invalid interface type and number
Router(config)#interface fa 0/0
%Invalid interface type and number
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#interface g0/0
%Invalid interface type and number
Router(config)#int g0/0
%Invalid interface type and number
Router(config)#interface g0/0/0
Router(config-if)#ip address 09.56.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

c

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
%Invalid interface type and number
Router(config)#interface Fa0/0
%Invalid interface type and number
Router(config)#interface Fa 0/0
%Invalid interface type and number
Router(config)#interface fa 0/0
%Invalid interface type and number
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#interface g0/0
%Invalid interface type and number
Router(config)#int g0/0
%Invalid interface type and number
Router(config)#interface g0/0/0
Router(config-if)#ip address 09.56.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface g0/0/1
Router(config-if)#ip address 09.56.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

c

```
Router#
*Oct 02, 20:49:49.4949: SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   9.56.1.1        YES manual up                    up
GigabitEthernet0/0/1   9.56.2.1        YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  down                  down
GigabitEthernet0/0/3   unassigned      YES unset  down                  down
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ip domain-name fasih.com
R1(config)#username admin privilege 15 secret cisco123
R1(config)#crypto key generate rsa
The name for the keys will be: R1.fasih.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Oct 2 20:53:23.15: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 15
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#exit
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 09.56.1.1

Pinging 09.56.1.1 with 32 bytes of data:

Reply from 9.56.1.1: bytes=32 time<1ms TTL=255
Reply from 9.56.1.1: bytes=32 time<1ms TTL=255
Reply from 9.56.1.1: bytes=32 time<1ms TTL=255
Reply from 9.56.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 9.56.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l admin 09.56.1.1

Password:



R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```

S

```
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g0/0/1
R1(config-if)#ip address 09.56.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#end
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   9.56.1.1        YES manual up                    up
GigabitEthernet0/0/1   9.56.2.1        YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  down                  down
GigabitEthernet0/0/3   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  up                    down
R1#
```

S

```
C:\>ping 09.56.1.1

Pinging 09.56.1.1 with 32 bytes of data:

Reply from 9.56.1.1: bytes=32 time<1ms TTL=255
Reply from 9.56.1.1: bytes=32 time<1ms TTL=255
Reply from 9.56.1.1: bytes=32 time<1ms TTL=255
Reply from 9.56.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 9.56.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 09.56.2.1

Pinging 09.56.2.1 with 32 bytes of data:

Reply from 9.56.2.1: bytes=32 time<1ms TTL=255
Reply from 9.56.2.1: bytes=32 time<1ms TTL=255
Reply from 9.56.2.1: bytes=32 time<1ms TTL=255
Reply from 9.56.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 9.56.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 09.56.2.20

Pinging 09.56.2.20 with 32 bytes of data:

Reply from 9.56.2.20: bytes=32 time<1ms TTL=127
Reply from 9.56.2.20: bytes=32 time<1ms TTL=127
Reply from 9.56.2.20: bytes=32 time<1ms TTL=127
Reply from 9.56.2.20: bytes=32 time<1ms TTL=127

Ping statistics for 9.56.2.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```